EXAM PAPERS PRACTICE

N3 Modelling with probability distributions A level only

Name: ______________________

Class: ______________________

Date: ______________________

Time: **502 minutes**

Marks: **412 marks**

Comments:

EXAM PAPERS PRACTICE
©2025 Exam Papers Practice. All rights reserved.

EXAM PAPERS PRACTICE

Q1.

Abu visits his local hardware store to buy six light bulbs.

He knows that 15% of all bulbs at this store are faulty.

(a) State a distribution which can be used to model the number of faulty bulbs he buys.

(1)

(b) Find the probability that all of the bulbs he buys are faulty.

(1)

(c) Find the probability that at least two of the bulbs he buys are faulty.

(2)

(d) Find the mean of the distribution stated in part (a).

(1)

(e) State two necessary assumptions in context so that the distribution stated in part (a) is valid.

(2)

(Total 7 marks)

Q2.

A survey of 120 adults found that the volume, X litres per person, of carbonated drinks they consumed in a week had the following results:

$$\sum x = 165.6 \qquad \sum x^2 = 261.8$$

(a) (i) Calculate the mean of X.

(1)

(ii) Calculate the standard deviation of X.

(2)

(b) Assuming that X can be modelled by a normal distribution find

(i) P(0.5 < X < 1.5)

(2)

(ii) P(X = 1)

(1)

(c) Determine with a reason, whether a normal distribution is suitable to model this data.

(2)

(d) It is known that the volume, Y litres per person, of energy drinks consumed in a week may be modelled by a normal distribution with standard deviation 0.21

For more help, please visit our website www.exampaperspractice.co.uk

EXAM PAPERS PRACTICE

Given that P(Y > 0.75) = 0.10, find the value of μ, correct to three significant figures.

(4)

(Total 12 marks)

Q3.

A survey has found that, of the 2400 households in Growmore, 1680 eat home-grown fruit and vegetables.

(a) Using the binomial distribution, find the probability that, out of a random sample of 25 households in Growmore, exactly 22 eat home-grown fruit and vegetables.

(2)

(b) Give a reason why you would **not** expect your calculation in part **(a)** to be valid for the 25 households in Gifford Terrace, a residential road in Growmore.

(1)

(Total 3 marks)

Q4.

During the 2006 Christmas holiday, John, a maths teacher, realised that he had fallen ill during 65% of the Christmas holidays since he had started teaching.

In January 2007, he increased his weekly exercise to try to improve his health.
For the next 7 years, he only fell ill during 2 Christmas holidays.

(a) Using a binomial distribution, investigate, at the 5% level of significance, whether there is evidence that John's rate of illness during the Christmas holidays had decreased since increasing his weekly exercise.

(6)

(b) State **two** assumptions, regarding illness during the Christmas holidays, that are necessary for the distribution you have used in part **(a)** to be valid.

For **each** assumption, comment, in context, on whether it is likely to be correct.

(4)

(Total 10 marks)

EXAM PAPERS PRACTICE
©2025 Exam Papers Practice. All rights reserved.

Q5.

A machine, which cuts bread dough for loaves, can be adjusted to cut dough to any specified set weight. For any set weight, μ grams, the actual weights of cut dough are known to be approximately normally distributed with a mean of μ grams and a fixed standard deviation of σ grams.

It is also known that the machine cuts dough to within 10 grams of any set weight.

(a) Estimate, with justification, a value for σ.

(2)

(b) The machine is set to cut dough to a weight of 415 grams.

As a training exercise, Sunita, the quality control manager, asked Dev, a recently

EXAM PAPERS PRACTICE

employed trainee, to record the weight of each of a random sample of 15 such pieces of dough selected from the machine's output. She then asked him to calculate the mean and the standard deviation of his 15 recorded weights.

Dev subsequently reported to Sunita that, for his sample, the mean was 391 grams and the standard deviation was 95.5 grams.

Advise Sunita on whether or not **each** of Dev's values is likely to be correct. Give numerical support for your answers.

(3)

(c) Maria, an experienced quality control officer, recorded the weight, y grams, of each of a random sample of 10 pieces of dough selected from the machine's output when it was set to cut dough to a weight of 820 grams. Her summarised results were as follows.

$$\sum y = 8210.0 \quad \text{and} \quad \sum (y - \bar{y})^2 = 110.00$$

Explain, with numerical justifications, why **both** of these values are likely to be correct.

(4)
(Total 9 marks)

Q6.

During June 2011, the volume, X litres, of unleaded petrol purchased per visit at a supermarket's filling station by private-car customers could be modelled by a normal distribution with a mean of 32 and a standard deviation of 10.

(a) Determine:

(i) $P(X < 40)$;

(ii) $P(X > 25)$;

(iii) $P(25 < X < 40)$.

(7)

(b) Given that during June 2011 unleaded petrol cost £1.34 per litre, calculate the probability that the unleaded petrol bill for a visit during June 2011 by a private-car customer exceeded £65.

(3)

(c) Give **two** reasons, in context, why the model $N(32, 10^2)$ is unlikely to be valid for a visit by **any** customer purchasing fuel at this filling station during June 2011.

(2)
(Total 12 marks)

Q7.

The records at a passport office show that, on average, 15 per cent of photographs that accompany applications for passport renewals are unusable.

EXAM PAPERS PRACTICE
©2025 Exam Papers Practice. All rights reserved.

Assume that exactly one photograph accompanies each application.

(a) Determine the probability that in a random sample of 40 applications:

(i) exactly 6 photographs are unusable;

(ii) at most 5 photographs are unusable;

(iii) more than 5 but fewer than 10 photographs are unusable.

(7)

(b) Calculate the mean and the standard deviation for the number of photographs that are unusable in a random sample of **32** applications.

(3)

(c) Mr Stickler processes 32 applications each day. His records for the previous 10 days show that the numbers of photographs that he deemed unusable were

8 6 10 7 9 7 8 9 6 7

By calculating the mean and the standard deviation of these values, comment, with reasons, on the suitability of the $\mathrm{B}(32, 0.15)$ model for the number of photographs deemed unusable each day by Mr Stickler.

(4)

(Total 14 marks)

Q8.

A bin contains a very large number of paper clips of different colours. The proportion of each colour is shown in the table.

Colour	White	Yellow	Green	Blue	Red	Purple
Proportion	0.15	0.15	0.20	0.15	0.22	0.13

(a) Packets are filled from the bin. Each filled packet contains exactly 30 paper clips which may be considered to be a random sample.

Use binomial distributions to determine the probability that a filled packet contains:

(i) exactly 2 purple paper clips;

(3)

(ii) a **total** of more than 10 red or purple paper clips;

(3)

(iii) at least 5 but at most 10 green paper clips.

(3)

(b) Jumbo packets are also filled from the bin. Each filled jumbo packet contains exactly 100 paper clips.

(i) Assuming that the number of paper clips in a jumbo packet may be considered

EXAM PAPERS PRACTICE

to be a random sample, calculate the mean and the variance of the number of **red** paper clips in a filled jumbo packet.

(2)

(ii) It is claimed that the proportion of red paper clips in the bin is greater than 0.22 and that jumbo packets do not contain random samples of paper clips.

An analysis of the number of red paper clips in each of a random sample of 50 filled jumbo packets resulted in a mean of 22.1 and a standard deviation of 4.17.

Comment, with numerical justification, on **each** of the two claims.

(3)

(Total 14 marks)

Q9.

During the winter, the probability that Barry's cat, Sylvester, chooses to stay outside all night is 0.35, and the cat's choice is independent from night to night.

(a) Determine the probability that, during a period of 2 weeks (14 nights) in winter, Sylvester chooses to stay outside:

(i) on at most 7 nights;

(2)

(ii) on at least 11 nights;

(2)

(iii) on more than 5 nights but fewer than 10 nights.

(3)

(b) Calculate the probability that, during a period of **3 weeks** in winter, Sylvester chooses to stay outside on exactly 4 nights.

(3)

EXAM PAPERS PRACTICE
©2025 Exam Papers Practice. All rights reserved.

(c) Barry claims that, during the summer, the number of nights per week, S, on which Sylvester chooses to stay outside can be modelled by a binomial distribution with $n = 7$ and $p = \frac{5}{7}$.

(i) Assuming that Barry's claim is correct, find the mean and the variance of S.

(2)

(ii) For a period of 13 weeks during the summer, the number of nights per week on which Sylvester chose to stay outside had a mean of 5 and a variance of 1.5.

Comment on Barry's claim.

(2)

(Total 14 marks)

For more help, please visit our website www.exampaperspractice.co.uk

EXAM PAPERS PRACTICE

Q10.

The proportion of passengers who use senior citizen bus passes to travel into a particular town on 'Park & Ride' buses between 9.30 am and 11.30 am on weekdays is 0.45.

It is proposed that, when there are n passengers on a bus, a suitable model for the number of passengers using senior citizen bus passes is the distribution B(n, 0.45).

(a) Assuming that this model applies to the 10.30 am weekday 'Park & Ride' bus into the town:

(i) calculate the probability that, when there are **16** passengers, exactly 3 of them are using senior citizen bus passes;

(3)

(ii) determine the probability that, when there are **25** passengers, fewer than 10 of them are using senior citizen bus passes;

(2)

(iii) determine the probability that, when there are **40** passengers, at least 15 but at most 20 of them are using senior citizen bus passes;

(3)

(iv) calculate the mean and the variance for the number of passengers using senior citizen bus passes when there are **50** passengers.

(2)

(b) (i) Give a reason why the proposed model may not be suitable.

(1)

(ii) Give a **different** reason why the proposed model would not be suitable for the number of passengers using senior citizen bus passes to travel into the town on the **7.15 am** weekday 'Park & Ride' bus.

(1)

EXAM PAPERS PRACTICE

(Total 12 marks)

©2025 Exam Papers Practice. All rights reserved.

Q11.

A survey of all the households on an estate is undertaken to provide information on the number of children per household.

The results, for the 99 households with children, are shown in the table.

Number of children (x)	1	2	3	4	5	6	7
Number of households (f)	14	35	25	13	9	2	1

(a) For these 99 households, calculate values for the mean and the standard deviation.

(3)

(b) In fact, 163 households were surveyed, of which 64 contained no children.

(i) For all 163 households, calculate the value for the mean number of children

EXAM PAPERS PRACTICE

per household.

(2)

(ii) State whether the value for the standard deviation, when calculated for all 163 households, will be smaller than, the same as, or greater than that calculated in part (a).

(1)

(iii) It is claimed that, for all 163 households on the estate, the number of children per household may be modelled approximately by a normal distribution.

Comment, with justification, on this claim. Your comment should refer to a fact other than the discrete nature of the data.

(2)

(Total 8 marks)

Q12.

A survey of all the households on an estate is undertaken to provide information on the number of children per household.

The results, for the 99 households with children, are shown in the table.

Number of children (x)	1	2	3	4	5	6	7
Number of households (f)	14	35	25	13	9	2	1

(a) For these 99 households, calculate values for:

(i) the median and the interquartile range;

(3)

(ii) the mean and the standard deviation.

(3)

EXAM PAPERS PRACTICE
©2025 Exam Papers Practice. All rights reserved.

(b) In fact, 163 households were surveyed, of which 64 contained no children.

(i) For all 163 households, calculate the value for the mean number of children per household.

(2)

(ii) State whether the value for the standard deviation, when calculated for all 163 households, will be smaller than, the same as, or greater than that calculated in part (a) (ii).

(1)

(iii) It is claimed that, for all 163 households on the estate, the number of children per household may be modelled approximately by a normal distribution.

Comment, with justification, on this claim. Your comment should refer to a fact other than the discrete nature of the data.

(2)

For more help, please visit our website www.exampaperspractice.co.uk

EXAM PAPERS PRACTICE

(Total 11 marks)

Q13.

Mr Alott and Miss Fewer work in a postal sorting office.

(a) The number of letters per batch, R, sorted incorrectly by Mr Alott when sorting batches of 50 letters may be modelled by the distribution B (50, 0.15).

Determine:

(i) P(R < 10);

(ii) P(5 ≤ R ≤ 10).

(4)

(b) It is assumed that the probability that Miss Fewer sorts a letter incorrectly is 0.06, and that her sorting of a letter incorrectly is independent from letter to letter.

(i) Calculate the probability that, when sorting a batch of **22** letters, Miss Fewer sorts exactly 2 letters incorrectly.

(3)

(ii) Calculate the probability that, when sorting a batch of **35** letters, Miss Fewer sorts at least 1 letter incorrectly.

(2)

(iii) Calculate the mean and the variance for the number of letters sorted **correctly** by Miss Fewer when she sorts a batch of **120** letters.

(2)

(iv) Miss Fewer sorts a random sample of 20 batches, each containing 120 letters. The number of letters sorted **correctly** per batch has a mean of 112.8 and a variance of 56.86.

Comment on the assumptions that the probability that Miss Fewer sorts a letter incorrectly is 0.06, and that her sorting of a letter incorrectly is independent from letter to letter.

(3)

(Total 14 marks)

Q14.

For the adult population of the UK, 35 per cent of men and 29 per cent of women do not wear glasses or contact lenses.

(a) Determine the probability that, in a random sample of 40 men, at most 15 do not wear glasses or contact lenses.

(3)

(b) Calculate the probability that, in a random sample of 10 women, exactly 3 do not wear glasses or contact lenses.

(3)

For more help, please visit our website www.exampaperspractice.co.uk

EXAM PAPERS PRACTICE

(c) (i) Calculate the mean and the variance for the number who **do** wear glasses or contact lenses in a random sample of 20 women.

(3)

(ii) The numbers wearing glasses or contact lenses in 10 groups, each of 20 women, had a mean of 16.5 and a variance of 2.50.

Comment on the claim that these 10 groups were **not** random samples.

(3)
(Total 12 marks)

Q15.

For the adult population of the UK, 35 per cent of men and 29 per cent of women do not wear glasses or contact lenses.

(a) Determine the probability that, in a random sample of 40 men:

(i) at most 15 do not wear glasses or contact lenses;

(3)

(ii) more than 10 but fewer than 20 do not wear glasses or contact lenses.

(3)

(b) Calculate the probability that, in a random sample of 10 women, exactly 3 do not wear glasses or contact lenses.

(3)

(c) (i) Calculate the mean and the variance for the number who **do** wear glasses or contact lenses in a random sample of 20 women.

(3)

(ii) The numbers wearing glasses or contact lenses in 10 groups, each of 20 women, had a mean of 16.5 and a variance of 2.50.

Comment on the claim that these 10 groups were **not** random samples.

(3)
(Total 15 marks)

EXAM PAPERS PRACTICE
©2025 Exam Papers Practice. All rights reserved.

Q16.

A hotel has 50 single rooms, 16 of which are on the ground floor. The hotel offers guests a choice of a full English breakfast, a continental breakfast or no breakfast. The probabilities of these choices being made are 0.45, 0.25 and 0.30 respectively. It may be assumed that the choice of breakfast is independent from guest to guest.

(a) On a particular morning there are 16 guests, each occupying a single room on the ground floor. Calculate the probability that exactly 5 of these guests require a full English breakfast.

(3)

(b) On a particular morning when there are 50 guests, each occupying a single room, determine the probability that:

For more help, please visit our website www.exampaperspractice.co.uk

EXAM PAPERS PRACTICE

(i) at most 12 of these guests require a continental breakfast;

(2)

(ii) more than 10 but fewer than 20 of these guests require no breakfast.

(3)

(c) When there are 40 guests, each occupying a single room, calculate the mean and the standard deviation for the number of these guests requiring breakfast.

(4)

(Total 12 marks)

Q17.

Each weekday, Monday to Friday, Trina catches a train from her local station. She claims that the probability that the train arrives on time at the station is 0.4, and that the train's arrival time is independent from day to day.

(a) Assuming her claims to be true, determine the probability that the train arrives on time at the station:

(i) on at most 3 days during a 2-week period (10 days);

(2)

(ii) on more than 10 days but fewer than 20 days during an 8-week period.

(3)

(b) (i) Assuming Trina's claims to be true, determine the mean and standard deviation for the number of times during a week (5 days) that the train arrives on time at the station.

(3)

(ii) Each week, for a period of 13 weeks, Trina's travelling colleague, Suzie, records the number of times that the train arrives on time at the station. Suzie's results are

2 2 4 1 2 3 3 2 2 0 3 2 0

Calculate the mean and standard deviation of these values.

(3)

(iii) Hence comment on the likely validity of Trina's claims.

(2)

(Total 13 marks)

Q18.

The table shows, for a particular population, the proportion of people in each of the four main blood groups.

Blood group	**O**	**A**	**B**	**AB**
Proportion	0.40	0.28	0.20	0.12

EXAM PAPERS PRACTICE
©2025 Exam Papers Practice. All rights reserved.

EXAM PAPERS PRACTICE

(a) A random sample of 20 people is selected from this population.

Determine the probability that the sample contains:

(i) at most 10 people with blood group O;

(2)

(ii) exactly 3 people with blood group A;

(3)

(iii) more than 4 but fewer than 8 people with blood group B.

(3)

(b) A random sample of 500 people is selected from this population.

Find values for the mean and variance of the number of people in the sample with blood group AB.

(2)
(Total 10 marks)

Q19.

Plastic clothes pegs are made in various colours.

The number of red pegs may be modelled by a binomial distribution with parameter p equal to 0.2.

The contents of packets of 50 pegs of mixed colours may be considered to be random samples.

(a) Determine the probability that a packet contains:

(i) less than or equal to 15 red pegs;

EXAM PAPERS PRACTICE
©2025 Exam Papers Practice. All rights reserved.

(2)

(ii) exactly 10 red pegs;

(2)

(iii) more than 5 but fewer than 15 red pegs.

(3)

(b) Sly, a student, claims to have counted the number of red pegs in each of 100 packets of 50 pegs. From his results the following values are calculated.

Mean number of red pegs per packet = 10.5

Variance of number of red pegs per packet = 20.41

Comment on the validity of Sly's claim.

(4)
(Total 11 marks)

Q20.

Kirk and Les regularly play each other at darts.

(a) The probability that Kirk wins any game is 0.3, and the outcome of each game is independent of the outcome of every other game.

Find the probability that, in a match of 15 games, Kirk wins:

(i) fewer than half of the games;

(4)

(ii) more than 2 but fewer than 7 games.

(3)

(b) Kirk attends darts coaching sessions for three months. He then claims that he has a probability of 0.4 of winning any game, and that the outcome of each game is independent of the outcome of every other game.

(i) Assuming this claim to be true, calculate the mean and standard deviation for the number of games won by Kirk in a match of 15 games.

(3)

(ii) To assess Kirk's claim, Les keeps a record of the number of games won by Kirk in a series of 10 matches, each of 15 games, with the following results:

8	5	6	3	9	12	4	2	6	5

Calculate the mean and standard deviation of these values.

(2)

(iii) Hence comment on the validity of Kirk's claim.

(3)

(Total 15 marks)

Q21.

Kirk and Les regularly play each other at darts.

(a) The probability that Kirk wins any game is 0.3, and the outcome of each game is independent of the outcome of every other game.

Find the probability that, in a match of 15 games, Kirk wins:

(i) exactly 5 games;

(3)

(i) fewer than half of the games;

(3)

(ii) more than 2 but fewer than 7 games.

(3)

(b) Kirk attends darts coaching sessions for three months. He then claims that he has a probability of 0.4 of winning any game, and that the outcome of each game is independent of the outcome of every other game.

EXAM PAPERS PRACTICE
©2025 Exam Papers Practice. All rights reserved.

EXAM PAPERS PRACTICE

(i) Assuming this claim to be true, calculate the mean and standard deviation for the number of games won by Kirk in a match of 15 games.

(3)

(ii) To assess Kirk's claim, Les keeps a record of the number of games won by Kirk in a series of 10 matches, each of 15 games, with the following results:

8 5 6 3 9 12 4 2 6 5

Calculate the mean and standard deviation of these values.

(2)

(iii) Hence comment on the validity of Kirk's claim.

(3)

(Total 17 marks)

Q22.

Each evening Aaron sets his alarm for 7 am. He believes that the probability that he wakes before his alarm rings each morning is 0.4, and is independent from morning to morning.

(a) Assuming that Aaron's belief is correct, determine the probability that, during a week (7 mornings), he wakes before his alarm rings:

(i) on 2 or fewer mornings;

(ii) on more than 1 but fewer than 5 mornings.

(5)

(b) Assuming that Aaron's belief is correct, calculate the probability that, during a 4-week period, he wakes before his alarm rings on exactly 7 mornings.

(3)

EXAM PAPERS PRACTICE

(c) Assuming that Aaron's belief is correct, calculate values for the mean and standard deviation of the number of mornings in a week when Aaron wakes before his alarm rings.

©2025 Exam Papers Practice. All rights reserved.

(2)

(d) During a 50-week period, Aaron records, each week, the number of mornings on which he wakes before his alarm rings. The results are as follows.

Number of mornings	0	1	2	3	4	5	6	7
Frequency	10	8	7	7	5	5	4	4

(i) Calculate the mean and standard deviation of these data.

(3)

(ii) State, giving reasons, whether your answers to part (d)(i) support Aaron's belief that the probability that he wakes before his alarm rings each morning is 0.4, and is independent from morning to morning.

(3)

EXAM PAPERS PRACTICE

(Total 16 marks)

Q23.

(a) The volume, X millilitres, of toothpaste in medium-sized tubes may be assumed to be normally distributed with a mean of 56 and a standard deviation of 2.5.

Determine the probability that the volume of toothpaste in a tube is:

(i) less than 60 ml;

(3)

(ii) between 50 ml and 60 ml;

(3)

(iii) exactly 55 ml.

(1)

(b) The volume, Y millilitres, of toothpaste in large-sized tubes may be assumed to be normally distributed with a standard deviation of 3.4.

Given that 98 per cent of these tubes contain more than 100 ml of toothpaste, determine the mean volume of toothpaste in a large-sized tube.

(4)

(Total 11 marks)

Q24.

On arrival at a business centre, all visitors are required to register at the reception desk. An analysis of the register, for a random sample of 100 days, results in the following information on the number, X, of visitors per day.

Number of visitors per day	**Number of days**
1– 10	13
11– 20	33
21– 25	17
26– 30	12
31– 35	8
36– 40	5
41– 50	5
51– 100	7
Total	**100**

For more help, please visit our website www.exampaperspractice.co.uk

EXAM PAPERS PRACTICE

(a) Calculate an estimate of:

(i) μ, the mean number of visitors per day;

(ii) σ, the standard deviation of the number of visitors per day.

(4)

(b) Give a reason, based upon the data provided, why X is **unlikely** to be normally distributed.

(1)

(c) (i) Give a reason why $\overline{X}$, the mean of a random sample of 100 observations on X, may be assumed to be normally distributed.

(1)

(ii) State, in terms of μ and σ, the mean and variance of $\overline{X}$.

(2)

(d) Hence construct a 99% confidence interval for μ.

(4)

(e) The receptionist claims that she registers on average more than 30 visitors per day, and frequently registers more than 50 visitors on any one day.

Comment on **each** of these **two** claims.

(4)
(Total 16 marks)

Q25.

The weight, X grams, of a particular variety of orange is normally distributed with mean 205 and standard deviation 25.

EXAM PAPERS PRACTICE

(a) Determine the probability that the weight of an orange is:

(i) less than 250 grams;

(3)

(ii) between 200 grams and 250 grams.

(3)

(b) A wholesaler decides to grade such oranges by weight. He decides that the smallest 30 per cent should be graded as small, the largest 20 per cent graded as large, and the remainder graded as medium.

Determine, to one decimal place, the maximum weight of an orange graded as:

(i) small;

(ii) medium.

(5)

EXAM PAPERS PRACTICE

(c) The weight, Y grams, of a second variety of orange is normally distributed with mean 175.

Given that 90 per cent of these oranges weigh less than 200 grams, calculate the standard deviation of their weights.

(4)
(Total 15 marks)

Q26.

(a) At a particular checkout in a supermarket, the probability that the barcode reader fails to read the barcode first time on any item is 0.07, and is independent from item to item.

(i) Calculate the probability that, from a shopping trolley containing 17 items, the reader fails to read the barcode first time on exactly 2 of the items.

(3)

(ii) Determine the probability that, from a shopping trolley containing 50 items, the reader fails to read the barcode first time on at most 5 of the items.

(2)

(b) At another checkout in the supermarket, the probability that a faulty barcode reader fails to read the barcode first time on any item is 0.55, and is independent from item to item.

Determine the probability that, from a shopping trolley containing 50 items, this reader fails to read the barcode first time on at least 30 of the items.

(3)

(c) At a third checkout in the supermarket, a record is kept of X, the number of times per 50 items that the barcode reader fails to read a barcode first time. An analysis of the records gives a mean of 10 and a standard deviation of 6.8.

EXAM PAPERS PRACTICE

(i) Estimate p, the probability that the barcode reader fails to read a barcode first time.

©2025 Exam Papers Practice. All rights reserved.

(1)

(ii) Using your estimate of p and assuming that X can be modelled by a binomial distribution, estimate the standard deviation of X.

(2)

(iii) Hence comment on the assumption that X can be modelled by a binomial distribution.

(2)
(Total 13 marks)

Q27.

The length, X centimetres, of eels in a river may be assumed to be normally distributed with mean 48 and standard deviation 8.

EXAM PAPERS PRACTICE

An angler catches an eel from the river. Determine the probability that the length of the eel is:

(a) exactly 60 cm;

(1)

(b) less than 60 cm;

(2)

(c) within 5% of the mean length.

(5)

(Total 8 marks)

Q28.

Jenny is practising for a clay-pigeon shooting competition. The probability that she scores a hit with any one shot is 0.8.

Assume that each shot is independent of every other shot.

(a) Write down the probability distribution for Y, the number of shots that Jenny takes at the target in order to score her first hit.

(1)

(b) Calculate the probability that Jenny first hits the target with her fifth attempt.

(2)

(c) Find values for the mean and variance of Y.

(2)

(Total 5 marks)

Q29.

A recent large-scale survey established that 15 per cent of cars have faulty brake lights.

(a) Calculate the probability that, in a random sample of 18 cars, exactly 2 cars have faulty brake lights.

(3)

(b) Determine the probability that, in a random sample of 50 cars, more than 5 cars but fewer than 10 cars have faulty brake lights.

(3)

(c) Find the probability that, in a random sample of 900 cars, at most 150 cars have faulty brake lights.

(5)

(d) At a set of traffic lights, a policewoman records the number, R, of **vehicles** with faulty brake lights, out of 50 successive vehicles stopping at the traffic lights.

Give a reason why the binomial distribution, B(50, 0.15), might **not** be an appropriate model for R.

EXAM PAPERS PRACTICE

(1)
(Total 12 marks)

Q30.

A manufacturer of balloons produces 40 per cent that are oval and 60 per cent that are round.

Packets of 20 balloons may be assumed to contain random samples of balloons. Determine the probability that such a packet contains:

(a) an equal number of oval balloons and round balloons;

(3)

(b) fewer oval balloons than round balloons.

(2)

A customer selects packets of 20 balloons at random from a large consignment until she finds a packet with exactly 12 round balloons.

(c) Give a reason why a binomial distribution is **not** an appropriate model for the number of packets selected.

(1)
(Total 6 marks)

Q31.

A chemist sells two types of bath cube. One type is relaxing and the other type is invigorating. The owner of a small hotel buys 50 cubes from the chemist. For each cube there is independently a probability of 0.40 that it will be relaxing.

(a) Find the probability that the 50 cubes will include:

(i) 20 or fewer relaxing cubes;

(3)

(ii) 20 or fewer invigorating cubes;

(3)

(iii) more relaxing cubes than invigorating cubes.

(2)

(b) The 50 cubes included 22 relaxing cubes. State, giving a reason, whether or not a binomial distribution will provide an appropriate model for the random variable, R, in **each** of the following cases.

(i) A guest at the hotel randomly selects one of the 50 cubes. If it is not a relaxing cube, he replaces it and again selects a cube at random. He continues this procedure until he obtains a relaxing cube. The random variable R denotes the number of cubes he selects until he obtains a relaxing cube.

(ii) The owner randomly selects 20 of the 50 cubes and places them in a bowl. The random variable R denotes the number of relaxing cubes placed in the

EXAM PAPERS PRACTICE
©2025 Exam Papers Practice. All rights reserved.

EXAM PAPERS PRACTICE

bowl.

(4)
(Total 12 marks)

Q32.

In an office a machine dispenses hot water for use in making tea, coffee or hot chocolate. The machine is filled with water at the beginning of each day. When 20 litres of water have been used the machine must be refilled.

(a) The daily demand for hot water from the machine may be modelled by a normal distribution with mean 26 litres and standard deviation 8 litres. Find the probability that, on a particular day, the machine:

(i) will not need to be refilled;

(ii) will need to be refilled exactly once.

(5)

(b) Find the probability that during a five-day week the demand for hot water from the machine will be less than 20 litres on each day;

(2)

(c) The machine is moved to a different office. Following the move, it is observed that the machine needs refilling at least once on 84% of days and needs refilling more than once on 12% of days. Assuming that the daily demand for hot water, following the move, may still be modelled by a normal distribution, find the mean and standard deviation.

(6)
(Total 13 marks)

Q33.

Eight friends take a picnic to a cricket match. As her contribution to the picnic, Hilda buys eight sandwiches at a supermarket. She selects the sandwiches at random from those on display. The probability that a sandwich is suitable for vegetarians is independently 0.3 for each sandwich.

(a) Find the probability that, of the eight sandwiches, the number suitable for vegetarians is:

(i) 2 or fewer;

(ii) exactly 2;

(iii) more than 3.

(7)

(b) Two of the eight friends are vegetarians. Hilda decides to ensure that the eight sandwiches she takes to the match will include at least two suitable for vegetarians. If, having selected eight sandwiches at random, she finds they include fewer than two suitable for vegetarians she will replace one, or if necessary two, of the sandwiches unsuitable for vegetarians with the appropriate number of sandwiches suitable for vegetarians.

EXAM PAPERS PRACTICE

State whether or not the binomial distribution provides an appropriate model for the number of sandwiches suitable for vegetarians which Hilda takes to the match. Explain your answer.

(2)

(c) In fact the eight sandwiches which Hilda took to the match contained four suitable for vegetarians. The first four friends to eat a sandwich were not vegetarians. Each selected one of the available sandwiches at random and ate it.

State whether or not the binomial distribution provides an appropriate model for the number of sandwiches suitable for vegetarians eaten by these four friends. Explain your answer.

(2)

(Total 11 marks)

Q34.

Gerhard walks to school each morning. The probability that he arrives late is 0.15 and is independent of whether he arrives late on any other morning.

(a) Find, for a week when he walks to school on five mornings:

(i) the probability he arrives late on two or fewer mornings;

(3)

(ii) the probability he arrives late on more than three mornings;

(2)

(iii) the mean and the standard deviation of the number of mornings on which he arrives late.

(3)

(b) The following table summarises the number of late arrivals of all pupils who attended Gerhard's school on five mornings of a particular week.

Number of late arrivals	Number of pupils
0	275
1	111
2	33
3	12
4	13
5	16

(i) Calculate the mean and the standard deviation of the data in the table.

(3)

EXAM PAPERS PRACTICE

(ii) Show that, for the pupils represented in the table, an estimate of the probability of a pupil arriving late on a particular morning is 0.15.

(1)

(c) It is suggested that the data in the table could be modelled by a binomial distribution.

(i) Comment on this suggestion in the light of your calculations in parts (a) and (b).

(2)

(ii) In the context of this question, give **two** possible reasons why the binomial distribution may **not** be a suitable model for the data in part (b).

(2)

(Total 16 marks)

Q35.

The weights, in grams, of the contents of tins of mackerel fillets are normally distributed with mean μ and standard deviation 2.5. The value of μ may be adjusted as required.

(a) Find the proportion of tins with contents weighing between 125.0 grams and 130.0 grams when μ = 129.0.

(5)

(b) (i) State, without proof, the value of μ which would maximise the proportion of tins with contents weighing between 125.0 grams and 130.0 grams.

(1)

(ii) Find the proportion of tins with contents weighing between 125.0 grams and 130.0 grams when μ is equal to the value you have specified in part (b)(i).

(3)

(c) Find, to one decimal place, the value of μ such that 99% of the tins have contents weighing more than 125.0 grams.

(4)

(Total 13 marks)

EXAM PAPERS PRACTICE
©2025 Exam Papers Practice. All rights reserved.